CSDBF: Dual-Branch Framework Based on Temporal–Spatial Joint Graph Attention With Complement Strategy for Hyperspectral Image Change Detection

Xianghai Wang, Keyun Zhao, Xiaoyang Zhao, and Siyao Li

***Abstract*— Hyperspectral image (HSI) change detection (CD) aims at obtaining internal components' change information of land cover and land use. In recent years, the development of convolutional neural networks (CNNs) has greatly promoted the research progress in this field. However, the fixed small-size convolution kernels used by CNNs have severely limited the receptive field of information. Another defect of most CNN-based models is their strong dependence on samples, and they are not competent for tasks with a small number of samples. Besides, the traditional CNN-based models can only perform convolution to learn the spatial–spectral features in the Euclidean space, which is not conducive to capturing the geometric changes in land covers in the HSIs. Differently, the graph attention network (GAT) has come into prominence due to its ability to capture the holistic topology structure of images flexibly, and the attention coefficients can be used to effectively model the long-range correlations between land covers. The semi-supervised nature of GAT is also well-suited to handle HSI-CD tasks with limited samples. Nevertheless, the pixel-level topology structure often generates expensive computational costs. To this end, a dual-branch framework based on temporal–spatial joint graph attention (TSJGAT) with complement strategy (CSDBF) is proposed for HSI-CD, which extracts superpixel- and pixel-level features from bitemporal HSIs in parallel and enables them to complement each other. The proposed CSDBF mainly consists of two branches: superpixel-level feature extraction branch (S-branch) and pixel-level feature extraction branch (P-branch). In the S-branch, we introduce the idea of GAT into HSI-CD for the first time and propose a novel TSJGAT module. Thus, the temporal–spatial features of HSIs are propagated and aggregated on the nonlinear graph structure, which makes the changed regions more discriminable. In the P-branch, pixel-level features are obtained by CNNs to correct uncertain factors caused by superpixel segmentation in the S-branch, which is complementary to the S-branch and lays a foundation for more accurate CD. Abundant experiments show that compared with other pioneer methods, the proposed CSDBF can improve the Kappa coefficient by more than 1.9% and 2.5% on average in general sampling rate situations and a low sampling rate situation, respectively, which shows better robustness and better detection accuracy than most existing state-of-the-art methods. The source code of this article can be downloaded from `https://github.com/zkylnnu/CSDBF`.**

***Index Terms*— Change detection (CD), convolutional neural network (CNN), graph attention network (GAT), hyperspectral image (HSI), superpixel.**

I. Introduction

With the continuous improvement of spectral imaging quality, hyperspectral images (HSIs) are increasingly widely used in Earth observation applications. Unlike other remote sensing images, HSIs are characterized by precise spectral sampling over a wide spectral wavelength range. Thus, HSIs with high spectral resolution provide the possibility to capture more sophisticated land cover dynamics [1]. HSI change detection (CD) aims to obtain the change information of the internal components of land cover using the time-series HSIs taken at the same geographic location. In recent years, HSI-CD has attracted more attention in agriculture and forestry detection, land cover renewal, urban change analysis, pollution detection, and other application fields [2], [3], [4], [5]. Certainly, it has become a hot topic in the field of remote sensing image information processing [6].

The early HSI-CD methods generally obtained the CD results through the generalized similarity measure between bitemporal HSIs, mainly including algebra-based methods, transformation-based methods, classification-based methods, statistical-modeling-based methods, unmixing-based methods [6], [7], [8], etc. The algebra-based methods, such as traditional difference, ratio, regression, and change vector analysis (CVA), determine the final changed area through threshold calculation based on similarity measurement [9]. The CD results of these methods depend on the band selection and prediction (PRE) bands to a large extent [10]. The transformation-based methods convert the images in the spatial domain into the feature domain to emphasize changed information and suppress unchanged information, such as principal component analysis (PCA) [11] and iteratively reweighted multivariate alteration detection (IR-MAD) [12]. Such methods work well in reducing the data redundancy between bands, but the spectral information suffers a certain loss. The classification-based methods provide a guarantee for obtaining " from–to" CD

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information by introducing a classification mechanism in CD. Common methods including support vector machine (SVM), minimum distance classifier, and maximum likelihood classifier (MLD) [13], [14] could effectively suppress the effect of noise interference on errors in different temporal HSIs. However, these methods mostly ignore the correlations between bitemporal HSIs, and the CD processes greatly depend on the adopted classification algorithm, resulting in poor robustness [15]. Methods based on statistical modeling mine the change attributes by statistically modeling the amplitude of multitemporal HSIs or their difference images [16], [17]. The key is how to obtain an accurate statistical model of difference images and estimate their parameters. At present, the probability density distribution of the multispectral images is mainly used as the prior distribution, and the research on the distribution characteristics of HSI pixels and differential image needs to be further studied. The unmixing-based methods are unique to HSI [18], [19], [20] and can effectively identify the components of land covers and their abundance in the pixels to a certain extent. Nevertheless, its availability greatly depends on the accuracy of spectral unmixing. For now, most of the widely used unmixing-based methods tend to focus on specific problems or depend on certain assumptions, which means weak generality.

Deep learning has attracted extensive attention from scholars in various fields for its powerful ability to represent learning recently and shows great potential in the field of CD [21]. In the preliminary stage of development, many CD methods with relatively simple network structures emerged, such as feature extraction methods based on stacked autoencoders (SAEs) [22], detection methods based on convolutional networks for biomedical image segmentation (U-NET) feature representation [23], and detection methods based on unsupervised feature learning [24]. Convolutional neural networks (CNNs), as a typical deep learning architecture, extract local features of images in the way of sliding windows through a fixed-size convolution kernel. Therefore, CNN with excellent characteristics such as invariable translation has attracted attention in the HSI-CD field. For example, Wang et al. [25] realized HSI-CD in an end-to-end way by integrating multiple subpixel information into a mixed affinity matrix. Zhan et al. [26] combined the spectral features with the spatial features and proposed a spectral–spatial CNN with Siamese (SSCNN-S) architecture, which transformed the CD problem into a measurement problem of similarity between two tensors. Ou et al. [27] presented a CNN framework named slow-fast band selection-feature fusion grouping network (SFBS-FFGNET), which used slow–fast band selection to reduce the number of bands to decrease the Hughes phenomenon and enhanced the orientation of fusion features by fusion feature grouping. Lin et al. [28] designed bilinear CNNs (BCNNs), which could take advantage of the relationships between input bitemporal images through a linear outer product operation. Although the above CD methods based on CNN have shown excellent performance to varying degrees, the characteristics of CNN determine the shortcomings of these methods as follows. On one hand, the deep convolution process is completed through the calculation of convolution kernels with fixed stack size and equal weight, which not only leads to the loss of edge information and limits the discriminant ability of the network but also hinders the exploration of long-range correlations [29], [30]; On the other hand, the traditional CNN methods can only extract features in the Euclidean space and are not considered as an optimal feature extractor for HSIs with complex land covers [31].

Recently, some researchers in the field of HSI processing have tried to alleviate the limitations of the receptive field of the CNN-based methods using the Transformer structure [32], [33], [34]. Among them, Wang et al. [34] proposed spectral-spatial-temporal-former (SST-Former), which is currently the only method using the Transformer structure to deal with the HSI-CD tasks as far as we know. In the SST-Former, a spectral transformer and a spatial transformer were designed to extract the spectral information and spatial information, respectively. Then, a temporal transformer was used to learn the change-related features and obtain the change rules between the objects of bitemporal HSIs. Certainly, the self-attention mechanism used by these Transformer-based methods enables them to capture the features with a larger receptive field. However, the large number of parameters of the Transformer-based methods makes them prone to overfitting, and the performance is often poor when the sampling rate is very low. Meanwhile, these methods often take a long time to train and test because of the huge amount of computation and low rate of convergence. Besides, most of the Transformer-based methods are weak in extracting the local spatial context information [35], [36].

In the past few years, graph neural networks (GNNs) based on feature extraction of irregular topology have been paid more attention in different research fields [37], [38]. Considering the importance of the propagation process for network training of GNNs, convolutional GNNs (ConvGNNs) generalize the plain convolution operation from fixed grid to topological graph structure, which generates an embedded representation of nodes by aggregating the features of nodes themselves and their neighbors [39]. ConvGNNs have both excellent characteristics as CNNs and GNNs. They are suitable for nodes and graphs of any topology and have shown great advantages in node classification, edge PRE, and other tasks. For the moment, ConvGNNs have been proved achievable in HSI classification and other fields. For detailed analysis, refer to Section II-A of this article.

Although lots of HSI-CD methods have been presented, this field still faces the following challenges on the whole: the first point is the lack of an effective HSI feature extraction framework in complex topological space to accurately capture the change information between multitemporal images; Second, the high-dimensional characteristic of HSIs makes it difficult to form a spatial pixel-level graph structure, and the use of a superpixel-level graph structure usually makes it difficult to identify the fine change information; Third, in the case of limited labeled samples, how to mine potential effective information from unlabeled samples to supplement the lack of labeled samples is also a key and difficult problem in this field. For these reasons, a dual-branch framework based on temporal–spatial joint graph attention (TSJGAT) with complement strategy is proposed in this article. The main contributions are as follows.

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1) A novel dual-branch feature extraction framework named CSDBF is proposed, which extracts the superpixel-level features and pixel-level features in parallel, and the advantages of two different feature extraction methods in the CD tasks are fully used to complement each other.

2) The idea of graph attention network (GAT) is introduced to HSI-CD for the first time, and we propose a novel TSJGAT network module. The proposed TSJGAT makes full use of the temporal–spatial joint correlations reflected by multitemporal HSIs and obtains a temporal–spatial affinity matrix so that the graph nodes can simultaneously aggregate multiple valid features according to the affinity matrix.

3) By combining the naive CNNs and the proposed TSJGAT in the proposed framework CSDBF, our network can capture both subtle and large-scale changes and shows great potential in the CD task with a low sampling rate. Abundant experiments show that the proposed CSDBF achieves competitive performance on the six datasets and surpasses most state-of-the-art methods in HSI-CD tasks with a low sampling rate.

The remainder of this article is organized as follows: Section II analyzes the related work, including the applications of GNNs in HSI processing, and the exploration of the temporal–spatial joint correlations of bitemporal HSIs; Section III describes the proposed HSI-CD method CSDBF, including the superpixel segmentation module, two important branches, feature fusion, and CD. Section IV presents the experimental results and visual analysis of this article, including plain sampling rate experiments, low sampling rate experiments, and ablation experiments; Section V provides the conclusion.

II. Related Work

A. Hyperspectra in GNN

GNN was first proposed by Scarselli et al. [40], [41] as a semi-supervised neural network architecture that can directly process the graph structure data. It plays an important role in solving problems such as nodes' classification, motion recognition, traffic PRE, and other problems with the topological structure. ConvGNNs are the most commonly used kinds of GNNs that aim at extending the convolution operation to the graph structure data to extract topological features effectively. Depending on the working domain, ConvGNNs fall into two categories: spectral-based and spatial-based.

On one hand, graph convolutional networks (GCNs) are typical of the spectral-based ConvGNNs, which were first proposed by Bruna et al. [42]. In this network, the convolution operation of the traditional Euclidean space is transferred to the graph network, and the Laplace matrix of the graph is used to generate the convolution operation of the graph. However, due to the high computational complexity of the Laplace matrix eigenvalue decomposition, this network architecture is usually only suitable for the calculation of small-scale graphs. As the graph convolution operation in [42] is global, the number of parameters is large and the computational complexity is high, and literature [43] extended it. A graph convolution structure with fast local convolution was proposed using polynomial expansion to approximate graph convolution and Chebyshev polynomial with iterative convolution kernel. Works based on [42] and [43], [44] enabled the GCN methods to be available in larger graph structures by designing a simple and outstanding hierarchical propagation rule, which effectively reduced the computational complexity and work on graph structure straightly. In the field of HSI processing, Hong et al. [29] successfully applied GCN to HSI classification and achieved better classification results than the traditional CNN methods. They constructed a graph structure at the pixel level and provided three different schemes for the fusion of CNN and GCN. Different from [29], Liu et al. constructed a graph structure at the superpixel level; superpixel-level features and pixel-level features were fused by the concat operation. Their work has also achieved excellent classification performance [45].

On the other hand, Veličković et al. [46] applied the self-attention mechanism to a graph structure and proposed a GAT, which was considered a typical representative of the spatial-based ConvGNNs. Unlike GCN which sets weights between graph nodes in advance, GAT obtains the attention coefficients between graph nodes by learning the correlations between them. In particular, all the attention coefficients in GAT are only related to the features of nodes, not the structure of the graph, so it has a stronger learning ability. Thus, Sha et al. [47] proposed a semi-supervised network based on GATs for HSI classification (S^2GATs), in which the attention mechanism was used to measure the spatial–spectral similarity between any neighbor nodes in the graph. Afterward, Xu et al. [48] presented a spectral–spatial residual GAT (S^2RGANet) for solving the same HSI classification problem. In this S^2RGANet, the discriminant spectral feature is captured by two spectral residual modules, and the local spatial residual information is adaptively aggregated from neighbor nodes. Besides, Qu et al. [49] proposed a GCN-based network named dual-branch difference amplification graph convolutional network (D^2AGAN), which was the first application of GCN in the HSI-CD field and obtained outstanding results with a low sampling rate.

As a novel framework for directly learning the graph structure data, GNN has achieved great success in the computer vision field by formulating certain strategies for graph nodes and edges to realize the standard representation of the graph structure data. Although GNN has attracted great attention in the HSI processing field in recent years, its applications are still few. Especially, GNN is still in its infancy in HSI-CD and many problems remain to be further studied.

B. Temporal–Spatial Joint Correlations

In the CD of multitemporal HSIs, the image sequence can be regarded as the geographical evolution process in the time dimension, so it has a potential indicative effect on the geographical distribution changes during closure time [50]. At present, CNN has been used to explore the temporal–spatial joint correlations in bitemporal HSIs. Song et al. [51] proposed a cross temporal interaction symmetric attention network (CSANet), which explored the temporal–spatial features between bitemporal HSIs through a self-attention mechanism to enhance the ability to recognize changes; Shi et al. [52]

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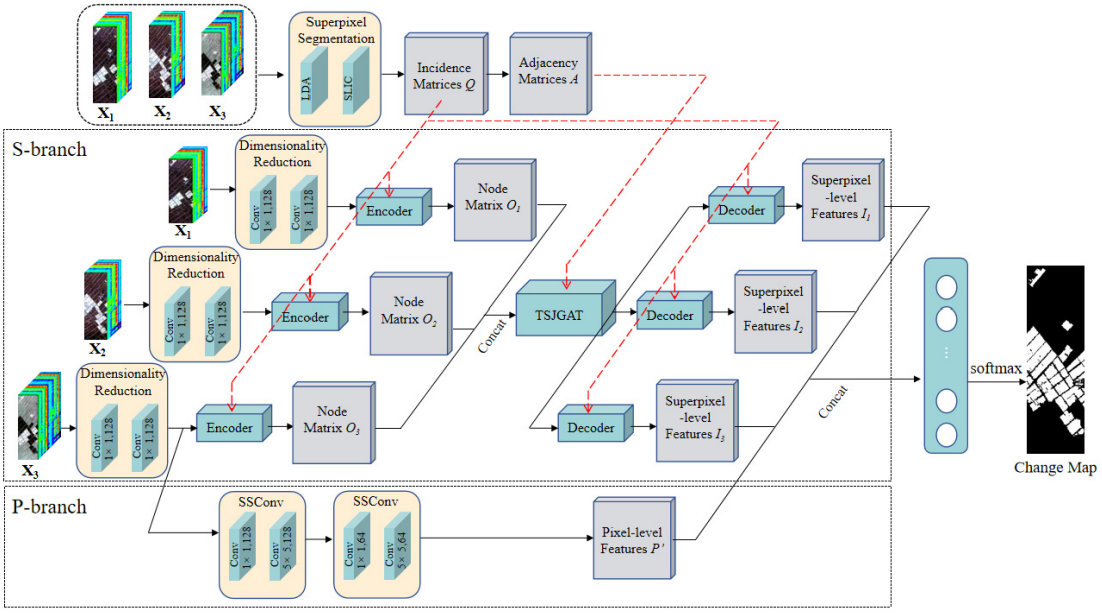


Fig. 1. Architecture of the proposed CSDBF.

proposed a multipath convolutional long short-term memory network (MP-ConvLSTM) to take advantage of LSTM and CNN. The MP-ConvLSTM adopts a Siamese CNN to extract the spatial–temporal features first, and models the temporal dependencies of bitemporal HSIs by convolutional LSTM for extracting joint multiscale temporal–spatial–spectral features. The aforementioned two studies have shown that the use of temporal–spatial joint correlations between bitemporal HSIs in the Euclidean space can effectively improve the ability of the network to discriminate real change information. To this end, the temporal–spatial joint correlations between bitemporal HSIs in the graph domain will be fully used in this article, and we expect them to play a vital role in this research field.

III. Proposed Method

We denote bitemporal HSIs as $\mathbf{X}_1$,$\mathbf{X}_2 \in \mathbb{R}^{H\times W\times B}$, and $\mathbf{V} \in \mathbb{R}^{H\times W}$ is the corresponding ground truth, where H and W denote the height and width of the spatial dimension of HSIs, respectively, and B denotes the number of bands of HSIs. Unlike most deep-learning-based HSI-CD methods that directly extract features from $\mathbf{X}_1$ and $\mathbf{X}_2$, we concatenate them by the channel dimension to obtain the third combined HSI $\mathbf{X}_3 \in \mathbb{R}^{H\times W\times 2B}$; then the three HSIs are further processed by the proposed CSDBF (see Fig. 1). As shown in Fig. 1, the proposed framework CSDBF is mainly composed of two parts: S-branch and P-branch. In addition, there is also some preprocessing work including the superpixel segmentation module. Linear discriminant analysis (LDA) and simple linear iterative clustering (SLIC) methods are adopted in the superpixel segmentation module to complete the segmentation of HSIs. This module generates the incidence matrices and adjacency matrices, which will be used by the encoders and decoders. In the S-branch, the three HSIs are first input into the dimensionality reduction module and the results are converted into node matrices by the encoders. Then, the node matrices are concatenated together and input into the proposed TSJGAT. The proposed TSJGAT can complete the feature propagation in the topology and obtain a graph feature matrix. Finally, the graph feature matrix is split and converted into superpixel-level features by the decoders. Unlike S-branch, the P-branch only contains two spectral–spatial convolution (SSCONV) modules, and the dimension reduction result of $\mathbf{X}_3$ is used as the input of the SSCONV module. Finally, the P-branch generates the pixel-level features and they are fused with the superpixel-level features together to accomplish the HSI-CD task.

A. Construction of Low-Dimensional Graph Structure Based on Superpixel Segmentation

The high-dimensional and redundant characteristics of HSI make the construction of pixel-level graph structures very complicated. To this end, in this article, HSIs are segmented into several disjoint superpixels according to the spatial similarity by a superpixel segmentation method, and then the corresponding spatial adjacency matrices and temporal adjacency matrices are constructed using superpixels as graph nodes to form a low-dimensional graph structure. Since the superpixel segmentation process and the corresponding low-dimensional graph construction process of $\mathbf{X}_1$, $\mathbf{X}_2$, and $\mathbf{X}_3$ are the same, we take an HSI as an example to illustrate the overall process, as shown in Fig. 2.

1) Superpixel Segmentation: This article adopts the same superpixel segmentation algorithm as [49]—SLIC method, which divides all the pixels into superpixels according to spatial similarity [53]. Specifically, to quickly construct an appropriate graph structure, LDA is applied to preprocess the HSI for dimensionality reduction, and then SLIC is applied to perform superpixel segmentation. In the process of superpixel segmentation, the scale of segmentation is denoted as c, which means the average number of pixels per superpixel, and thus, the number of superpixels is $K = N/c$, where $N = H \times W$ denotes the number of pixels in each band of HSI. The

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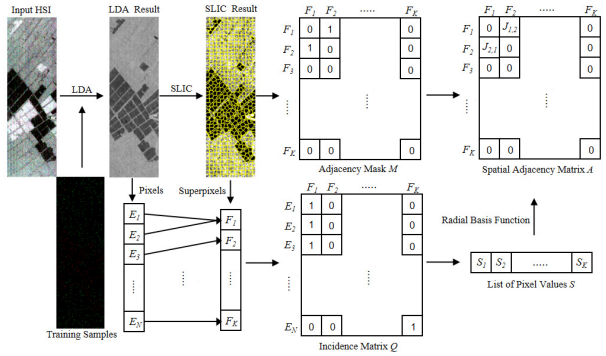


Fig. 2. Flowchart of the low-dimensional graph structure construction.

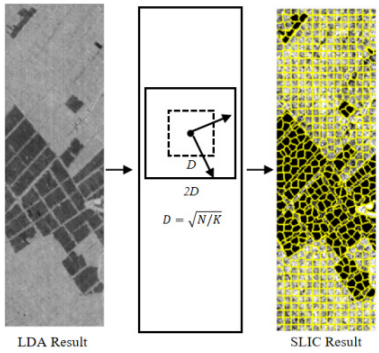


Fig. 3. Process of the SLIC method.

operation procedure of the SLIC method is as follows: First, K initial cluster centers are obtained by grid sampling at D intervals, where $D = (N/K)^{1/2}$; then, to avoid placing the centers of superpixels at class boundaries, new locations with lowest gradients are selected in their 3×3 neighborhood; finally, a k-means algorithm is applied to iteratively expand the local clusters and the segmentation process is completed. See Fig. 3 for the process of the SLIC method.

2) Construction of Temporal–Spatial Adjacency Matrix: For an HSI that is segmented into K superpixels, we denote the superpixel set as $F = \{F_i\}_{i=1}^{K}$, where $F_i = \{x_j^i\}_{j=1}^{Y_i}$ denotes the ith superpixel, x_j^i denotes the jth pixel in F_i, and Y_i is the number of pixels in F_i. Meanwhile, the result after dimensionality reduction by LDA is flattened by the spectral dimension to obtain the pixel set $E = \{E_i\}_{i=1}^{N}$, where E_i is the ith pixel in E.

First, to propagate the features between the pixel level and the superpixel level, inspired by [54], we define $Q \in \mathbb{R}^{N \times K}$ as the incidence matrix between pixels and superpixels, which is defined as follows:

$$Q_{i,j} = \begin{cases} 1, & \text{if } E_i \in F_j \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

where $Q_{i,j}(i = 1, 2, \ldots, N;\ j = 1, 2, \ldots, K)$ is the value of Q at spatial location (i, j). The physical meaning of this formula is: If the pixel E_i is assigned to superpixel F_j, the value of the incidence matrix Q at location (i, j) is 1; otherwise, it is 0. Through the incidence matrix Q, we can encode pixel-level features into a graph node matrix, where each node represents a superpixel. Furthermore, to ensure that each node in the TSJGAT module only aggregates the features of its related nodes, we need to obtain the spatial adjacency matrices and the temporal adjacency matrices. Then we can constrain the feature propagation on the graph structure according to the weights between nodes in the adjacency matrices.

Next, based on the contiguity relationship between superpixels, the adjacency mask $M \in \mathbb{R}^{K \times K}$ can be obtained as follows:

$$M_{i,j} = \begin{cases} 1, & \text{if } F_i \text{ and } F_j \text{ are adjacent} \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

where $M_{i,j}(i = 1, 2, \ldots, K;\ j = 1, 2, \ldots, K)$ denotes the value of M at location (i, j).

Then, we can calculate the average of the pixels contained in each superpixel according to the incidence matrix Q and use it as the pixel value for that superpixel. Thus, a pixel value list of superpixels $S \in \mathbb{R}^{K}$ is obtained, which is defined as follows:

$$S_i = \frac{1}{K_i} \sum_{j=1}^{K_i} x_j^i \tag{3}$$

where $S_i(i = 1, 2, \ldots, K)$ denotes the ith element of S, K_i is the number of pixels contained in the ith superpixel, and x_j^i is the jth pixel in the ith superpixel.

a) Construction of spatial adjacency matrices: We can construct the spatial adjacency matrix $A \in \mathbb{R}^{K \times K}$ through the adjacency mask M and the pixel value list S

$$A_{i,j} = \begin{cases} J_{i,j}, & \text{if } M_{i,j} > 0 \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

where $A_{i,j}(i = 1, 2, \ldots, K;\ j = 1, 2, \ldots, K)$ denotes the value of A at location (i, j), and $J_{i,j}$ denotes the similarity between superpixels, which is obtained by the radial basis function

$$J_{i,j} = \exp\left(-\frac{|S_i - S_j|^2}{\sigma^2}\right) \tag{5}$$

where S_i and S_j denote the ith element and the jth element of S, respectively. σ is a free parameter that we set to ten in our experiments.

So far, we have constructed a spatial adjacency matrix of superpixels in the HSIs. Repeat the above process, we can construct three spatial adjacency matrices, which correspond to $\mathbf{X}_1$, $\mathbf{X}_2$, and $\mathbf{X}_3$. They are defined as $\{A^{i,i} \mid i = 1, 2, 3\}$.

b) Construction of temporal adjacency matrices: We can define the temporal adjacency matrix between HSIs $\mathbf{X}_p$ and $\mathbf{X}_q$ as $\tilde{A}^{p,q} \in \mathbb{R}^{K^p \times K^q}$, where K^p and $K^q (p \neq q)$ denote the number of superpixels of different temporal HSIs. The temporal adjacency matrix can be calculated as

$$\tilde{A}_{i,j}^{p,q} = \exp\left(-\frac{|S_i^p - S_j^q|^2}{\sigma^2}\right) \tag{6}$$

where $\tilde{A}_{i,j}^{p,q}(i = 1, 2, \ldots, K^p;\ j = 1, 2, \ldots, K^q)$ denotes the value of $\tilde{A}^{p,q}$ at location (i, j), S_i^p denotes the ith element in the pixel value list of $\mathbf{X}_p$, S_j^q denotes the jth element in the

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pixel value list of $\mathbf{X}_q$, and σ is the same as (5). Furthermore, the adjacency matrix $\tilde{A}^{p,q}$ is normalized by row as follows:

$$A_{i,j}^{p,q} = \frac{\tilde{A}_{i,j}^{p,q} - g_i}{\left(\sum_{k=1}^{K^q}\left(\tilde{A}_{i,k}^{p,q} - g_i\right)^2\right)/K^q} \tag{7}$$

where $g_i = (\sum_{k=1}^{K^q} \tilde{A}_{i,k}^{p,q})/K^q$ is the mean value of the elements in the ith row of the adjacency matrix $\tilde{A}^{p,q}$, and $A_{i,j}^{p,q}$ denotes the value of the normalized temporal adjacency matrix at location (i, j). It can be seen from the above normalization process that the relationship between nodes in the temporal adjacency matrix has directionality, which is different from the spatial adjacency matrix. In the same way, the temporal adjacency matrices between $\mathbf{X}_1$, $\mathbf{X}_2$, and $\mathbf{X}_3$ can be constructed according to the above process, and we define them as $\{A^{i,j} \mid i = 1, 2, 3; j = 1, 2, 3; i \neq j\}$. Through the temporal adjacency matrices and the correlation threshold set in the TSJGAT module, we can ensure that the nodes of $\mathbf{X}_p$ aggregate only the features of the nodes of $\mathbf{X}_q$ that are strongly associated with them.

B. TSJGAT Module and Superpixel-Level Feature Extraction

1) Preprocessing: For enhancing the robustness and discrimination of deep spectral feature learning and avoiding the Hughes phenomenon of HSIs, $\mathbf{X}_1$, $\mathbf{X}_2$, and $\mathbf{X}_3$ are first processed by two layers of naive 1×1 convolution to reduce the dimension (see the dimensionality reduction module in Fig. 1). Then three new feature maps $X_1', X_2', X_3' \in \mathbb{R}^{H \times W \times C}$ are obtained, where C denotes the new number of bands and $C < B$. Therefore, the feature maps X_1', X_2', X_3' are transformed from the Euclidean structure to the graph structure by the encoders, and the specific process is as follows.

a) Encoding: First, the 3-D feature maps X_1', X_2', X_3' are flattened by the spatial dimension, and the corresponding 2-D feature maps $I_1', I_2', I_3' \in \mathbb{R}^{N \times C}$ are obtained. Then we encode the pixel-level features as graph node matrices $O_n \in \mathbb{R}^{K^n \times C}$ ($n = 1, 2, 3$; K^n is the number of superpixels in X_n') according to (8)

$$O_n = \widehat{Q}_n^T \times I_n' \tag{8}$$

where $\widehat{Q}_n$ denotes the normalized Q_n by column (see (1)), and $\widehat{Q}_n^T$ denotes the transpose of $\widehat{Q}_n$. Then we have

$$\begin{cases} O_1 = [h_1, h_2, \ldots, h_{K^1}]^T \\ O_2 = \left[h_{K^1+1}, h_{K^1+2}, \ldots, h_{K^1+K^2}\right]^T \\ O_3 = \left[h_{K^1+K^2+1}, h_{K^1+K^2+2}, \ldots, h_{K^1+K^2+K^3}\right]^T. \end{cases} \tag{9}$$

b) Concat: Furthermore, we concatenate the three graph node matrices $O_n (n = 1, 2, 3)$ to form the following joint graph node matrix $h_{(K^1+K^2+K^3)\times C}$:

$$\begin{aligned} h = \big[&h_1, h_2, \ldots, h_{K^1}, h_{K^1+1}, h_{K^1+2}, \ldots, h_{K^1+K^2} \\ &h_{K^1+K^2+1}, h_{K^1+K^2+2}, \ldots, h_{K^1+K^2+K^3}\big] \end{aligned} \tag{10}$$

where $h_i \in \mathbb{R}^C$, $i = 1, 2, \ldots, K^1 + K^2 + K^3$.

2) Proposed TSJGAT Module: The proposed TSJGAT module takes the joint graph node matrix h as input (see Fig. 4), and the details are as follows:

a) Parameterization: We first define a learnable transformation $W \in \mathbb{R}^{C' \times C}$ to transform the joint graph node matrix h, where C' denotes the number of features contained in each node that is output by these graph attention layers, and the initialized parameters in W are set to 1.4. The graph node matrix after transformation is defined as $Wh_{C' \times (K^1+K^2+K^3)}$

$$Wh = W \times h^T. \tag{11}$$

b) Self-attention: To get the attention coefficients between different nodes, we adopt a shared self-attention mechanism. First, weight matrices $a_1, a_2 \in \mathbb{R}^{C'}$ are defined to parameterize the node matrix Wh (the initialized parameters are set to 1.4), as shown in Fig. 4(a). Then two groups of eigenvalues $f_1, f_2 \in \mathbb{R}^{K^1+K^2+K^3}$ are obtained according to (12)

$$f_i = Wh \times a_i \tag{12}$$

where $i = 1, 2$. Furthermore, by performing the self-attention mechanism to f_1 and f_2, we can obtain the attention coefficient matrix $e \in \mathbb{R}^{(K^1+K^2+K^3)\times(K^1+K^2+K^3)}$ between different nodes, as shown in Fig. 4(b)

$$e = \text{LeakyReLU}\left(f_1 + f_2^T\right) \tag{13}$$

where each element $e_{i,j} (i, j = 1, 2, \ldots, K^1 + K^2 + K^3)$ denotes the importance of graph node j to graph node i, and LeakyReLU$(\cdot)$ denotes the activation function LeakyReLU.

c) Correlation injection: The unified clustering strategies used by the traditional attention networks always enable a node to aggregate features of all other nodes. Differently, the cluster strategies adopted by the proposed TSJGAT vary according to the different temporal HSIs that the nodes belong to. That is, for the nodes belonging to the same temporal HSI, the current node and its first-order adjacent node are aggregated. For the nodes belonging to different temporal HSIs, the current node aggregates features of all the nodes with strong correlations to the current node in another temporal HSI. To this end, the attention coefficient matrix e is divided into nine areas $\{Z^{i,j} \mid i = 1, 2, 3; j = 1, 2, 3\}$, as shown in the attention coefficients matrix in Fig. 4. For the attention coefficients in the spatial attention areas $\{Z^{i,i} \mid i = 1, 2, 3\}$, we complete the correlation injection process according to the spatial adjacency matrices $\{A^{i,i} \mid i = 1, 2, 3\}$ (see Section III-A), which means a node is permitted to aggregate features of its one-order neighbors. For the attention coefficients in the temporal attention areas $\{Z^{i,j} \mid i = 1, 2, 3; j = 1, 2, 3; i \neq j\}$, the correlation injection process is completed according to the temporal adjacency matrices $\{A^{i,j} \mid i = 1, 2, 3; j = 1, 2, 3; i \neq j\}$, which means a node is going to aggregate the features of all other nodes that are strongly related to the current node. The specific implementation process is as follows:

First, the spatial adjacency matrices and temporal adjacency matrices are concatenated according to the aforementioned correspondence between different areas in the attention coefficient matrix e and different adjacency matrices. The following temporal–spatial joint adjacency matrix

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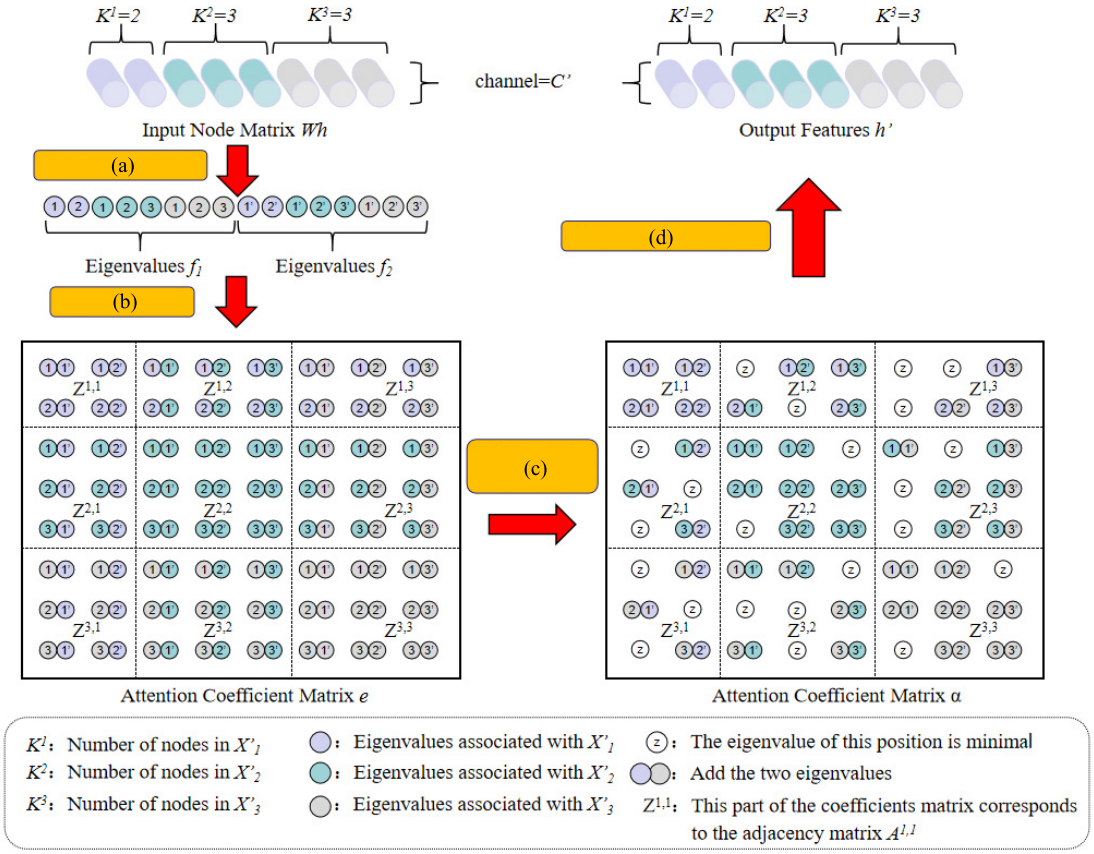


Fig. 4. Flowchart of the proposed TSJGAT (take the numbers of nodes $K^1 = 2$, $K^2 = 3$, and $K^3 = 3$ for example).

$A'_{(K^1+K^2+K^3)\times(K^1+K^2+K^3)}$ is obtained:

$$A' = \begin{bmatrix} A^{1,1} & A^{1,2} & A^{1,3} \\ A^{2,1} & A^{2,2} & A^{2,3} \\ A^{3,1} & A^{3,2} & A^{3,3} \end{bmatrix}. \tag{14}$$

Then, we finish the correlation injection of the attention coefficient matrix e according to the temporal–spatial joint adjacency matrix A', as shown in Fig. 4(c), and a new attention coefficient matrix α is obtained, that is,

$$\alpha_{i,j} = \begin{cases} e_{i,j}, & \text{if } A'_{i,j} > 0 \\ \min, & \text{otherwise} \end{cases} \tag{15}$$

where $\alpha_{i,j}$ is the value of α at the location (i, j), $A'_{i,j}$ is the value of A' at the location (i, j), and min is a minimum and is set to 9e-15 in our experiments.

d) Feature propagation: Finally, the softmax activation function is used to normalize α, and a normalized attention coefficient matrix $\beta_{(K^1+K^2+K^3)\times(K^1+K^2+K^3)}$ is obtained and used to complete feature aggregation of Wh [see formula (11)] with the exponential linear units (ELUs) activation function together. Therefore, the process of getting the output features $h'_{(K^1+K^2+K^3)\times C}$ of nodes can be written as follows:

$$\begin{cases} \beta = \text{softmax}(\alpha) \\ h'_i = \text{ELU}\left(\sum_{j=1}^{K^1+K^2+K^3} \left(\beta_{i,j} \times Wh_j\right)\right) \end{cases} \tag{16}$$

where $\beta_{i,j}(i, j = 1, 2, \ldots, K^1 + K^2 + K^3)$ is the value of β at the location (i, j), Wh_j denotes the elements in the jth column of Wh, and h'_i is the ith row vector of the output feature h'.

e) Multihead attention: To better stabilize the learning process of the TSJGAT module, this article extends the above TSJGAT to use multihead attention, G independent graph attention processes are adopted (G is set to 3 in our experiments), and the features obtained by each graph attention process are concatenated to form the final feature matrix $h''_{(K^1+K^2+K^3)\times 3C}$

$$h''_i = \text{Concat}\,|_{m=1}^{G}\left(\sum_{j=1}^{K^1+K^2+K^3}\left(\beta_{i,j}^{m} \times W^m h_j\right)\right) \tag{17}$$

where Concat denotes the concat operation along the spectral dimension, h''_i is the ith row vector of the final output feature h'', $\beta_{i,j}^{m}$ denotes the attention coefficient matrix of the mth graph attention process, and W^m is the corresponding weight matrix of the mth graph attention process.

f) Decoding: First, we separated the final feature matrix h'' according to the way we concatenate the node matrices $O_n(n = 1, 2, 3)$ in the preprocessing of TSJGAT. The separated node matrices can be defined as $O'_n \in \mathbb{R}^{K^n\times 3C}(i = 1, 2, 3)$, where K^n denotes the number of superpixels in X'_n. Then, the spilt node matrices are decoded, that is,

$$I_n = Q^n \times O'_n \tag{18}$$

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where Q^n is the incidence matrix between pixels and superpixels [see (1)], and I_n denotes the obtained pixel-level features after decoding.

The superpixel-level features can be restored into pixel-level features by the decoding process, and the transformation from the topological structure to the original Euclidean structure is achieved.

C. Pixel-Level Feature Extraction

The aforementioned TSJGAT effectively uses the temporal–spatial joint correlations of bitemporal HSIs to accomplish the feature propagation in irregular topological structures, and the extracted superpixel-level features with good separability lay a foundation for effective PRE of regional categories. Considering that the superpixel-level feature extraction branch may cause error accumulation due to superpixel segmentation, another pixel-level feature extraction branch is designed in this article to improve the insufficient PRE of the S-branch for discrete changed regions. To extract the spatial and spectral features of the images simultaneously, this article uses the SSCONV proposed in [55] to extract the pixel-level features from X_3' (see Fig. 1). Specifically, in SSCONV, a 1-D kernel and a 2-D kernel are used to replace the 3-D convolutional kernel. They can extract the spectral features and spatial features, respectively. Compared with the direct use of a 3-D kernel, the computation complexity is greatly reduced and the robustness is enhanced. The pixel-level feature extraction process can be defined as

$$P_i = \text{LeakyReLU}\left(u_j^i * \text{LeakyReLU}\left(y_j^i * \widetilde{P_i}\right) + b_j^i\right) \tag{19}$$

where $\widetilde{P_i} = BN(P_{i-1})$ denotes the normalized output data P_{i-1} of the $(i\text{-}1)$th convolution layer, and $P_{i-1} = X_3'$ when $i = 1$. u_j^i, y_j^i, and b_j^i denote the jth spatial 2-D kernel, spectral 1-D kernel, and bias of the ith layer, respectively. Through this branch, the final pixel-level feature P' is obtained.

D. Feature Fusion and CD

So far, the superpixel-level features $I_i (i = 1, 2, 3)$ and pixel-level features P' have been extracted by two branches. Then, fusion features are formed by the concat operation to make the two features complement each other and provide effective discrimination for CD tasks together. Thus, the fusion feature U is defined as follows:

$$U = \text{Concat}\left(I_1, I_2, I_3, P'\right). \tag{20}$$

Furthermore, fusion feature U is entered into the final fully connected layer to aggregate change information and separate unchanged information. Certainly, a softmax activation function is applied after the fully connected layer to calculate the probability that a pixel belongs to a certain class, which judges whether the pixel has changed and obtains the final PRE change map $\widehat{V} \in \mathbb{R}^{HW\times 2}$. Moreover, since a binary CD task is a binary classification problem in nature, the normal binary cross-entropy (BCE) loss function is applied to train the proposed CSDBF

$$L_{\text{BCE}} = -\sum_{p=1}^{N}\left(V_p' \log\left(\widehat{V}_p\right) + \left(1 - V_p'\right)\log\left(1 - \widehat{V}_p\right)\right) \tag{21}$$

where V' and $\widehat{V}$ denote the ground-truth map after one-hot encoding and the change map predicted by CSDBF, respectively. p denotes a location in HSIs. The more accurate the PRE, the smaller the value of L_{BCE}. Thus, in the course of network training, we optimize it by continuously decreasing the value of L_{BCE}.

IV. Experiments and Analysis

For the reason that it is very difficult to obtain the ground-truth maps of the HSI-CD tasks, HSI-CD tasks based on low sampling rates have attracted a lot of attention. To verify the effectiveness of the proposed framework CSDBF, two groups of experiments are carried out: The first group is the CD experiments with plain sampling rates; another group is the CD experiments with a low sampling rate. Meanwhile, the ablation experiments of different branches of CSDBF are carried out. For the sake of evaluating the CD performance of different methods, the commonly used evaluation indices overall accuracy (OA) and Kappa coefficient are adopted in this article, and the calculation formulas are as follows:

$$OA = \frac{TP + TN}{TP + TN + FP + FN} \tag{22}$$

$$\text{Kappa} = \frac{OA - \text{PRE}}{1 - \text{PRE}} \tag{23}$$

where true positives (TPs), true negatives (TNs), false positives (FPs), and false negatives (FNs) denote the correctly predicted numbers of changed pixels and unchanged pixels, and wrongly predicted numbers of unchanged pixels and changed pixels, respectively. Meanwhile, the Kappa coefficient can measure the PRE accuracy from a consistency point of view, while OA only measures according to the ratio of the correct quantity to the total quantity. The calculation formula of PRE is as follows:

$$\text{PRE} = \frac{(TP + FP)(TP + FN) + (TN + FP)(TN + FN)}{(TP + TN + FP + FN)^2}. \tag{24}$$

The OA and Kappa metrics reflect the performance of the CD methods from different aspects according to the relationship between CD results and the ground-truth maps. In general, the larger the values, the better the CD performance. Besides, for datasets with severe class imbalance, the OA metric has a certain degree of deviation in the evaluation of the performance of the CD methods, while the Kappa metric can compensate for it by reflecting the consistency of detection.

The proposed method is carried out with Python-3.6 and Pytorch-1.9.0, on RTX-3090 graphic process unit (GPU). The learning rate is set to 0.0001 and the maximum number of epochs in the training process is set to 1000. The negative slope of the activation function LeakyReLU is set to 0.2. Besides, the methods with patchwise inputs often cut the image into patches and divide the training and testing samples

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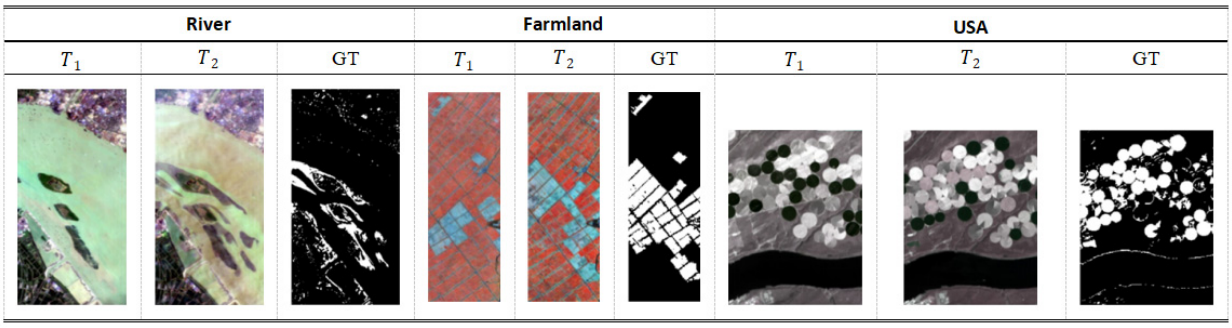


Fig. 5. Pseudocolor images and ground-truth images of the datasets in the first group of experiments.

TABLE I
DATASET INFORMATION

Dataset	River	Farmland	USA
Size	463×241	450×140	307×241
Spectral Bands	198	155	154
Date T_1	May 3rd, 2013	May 3rd,2006	May 1st,2004
Date T_2	Dec 31st, 2013	Apr 23rd,2007	May 8th,2007
Sensor	Hyperion	Hyperion	Hyperion

before sending them into the network. Differently, the position of labeled samples is marked as 1 in the entire image by the methods with imagewise inputs, and we distinguish the training samples from the testing samples by these masks. During the training and testing phases, the inputs and outputs of the CSDBF are imagewise.

A. First Group of Experiments: Plain Sampling Rate

1) Dataset Description: This group of experiments is conducted on River [25], Farmland [56], and USA [8] three public real-world bitemporal HSI-CD datasets. The detailed information of the three datasets is shown in Table I. Fig. 5 shows the pseudocolor images and ground-truth images. White and black in the ground-truth images represent the changed and unchanged pixels, respectively (the CD results of different methods are also represented in this way).

2) Sampling Rate and Compared Methods: This group of experiments adopts the same sampling rate as most literature [25], [26], [27], that is, selecting 3.36%, 20.95%, and 9.77% of samples for training on River, Farmland, and USA datasets. About 1% of the samples are used for validation. Especially, for the compared methods, we adopt the sampling mode selected in their articles which is sampling by the ratio of positive samples to negative samples (1:2 on the River and Farmland datasets and about 1:1.2 on the USA dataset), while all the training samples were completely randomly selected for the proposed CSDBF. Besides, in our experiments, the superpixel segmentation scales of River, Farmland, and USA are 25, 10, and 15, respectively. Some widely used and advanced approaches are selected for comparative analysis in experiments, including patch-based CNN (PBCNN) [57], general end-to-end two-dimensional cnn framework (GETNET) [25], deep slow feature analysis (DSFA) [58], hybrid spectral CNN (HybridSN) [59], double-branch dual-attention (DBDA) mechanism network [60], and recent methods SSCNN-S [26], SFBS-FFGNET [27].

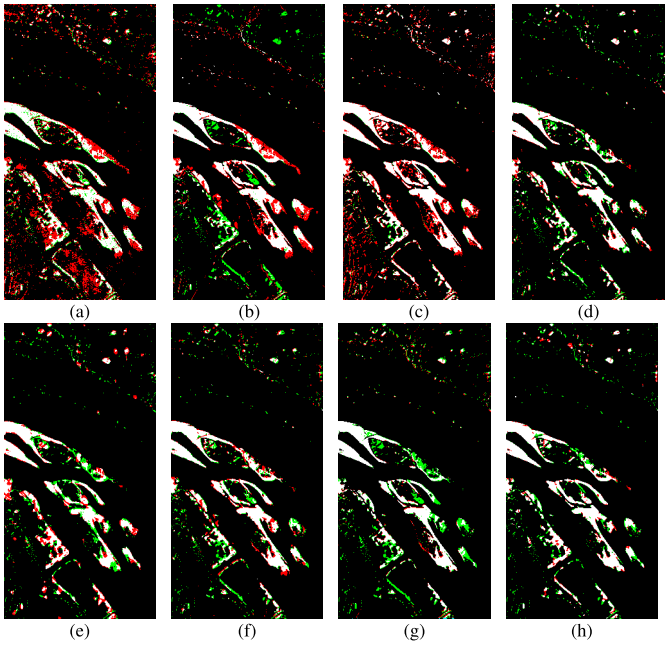


Fig. 6. CD results of different methods on the River dataset (the missed alarms and false alarms are marked in green and red, respectively). (a) PBCNN. (b) DSFA. (c) GETNET. (d) HybridSN. (e) DBDA. (f) SSCNN-S. (g) SFBS-FFGNET. (h) CSDBF.

TABLE II
QUANTITATIVE CD EVALUATIONS ON THE RIVER DATASET

Method	OA	Kappa
PBCNN	0.9139	0.5585
DSFA	0.9482	0.7207
GETNET	0.9499	0.7472
HybridSN	0.9631	0.7608
DBDA	0.9596	0.7260
SSCNN-S	0.9640	0.7431
SFBS-FFGNET	0.9670	0.7710
CSDBF	**0.9697±0.0019**	**0.7982±0.0144**
	↑0.27%	**↑2.72%**

3) Analysis of Experimental Results:

a) River dataset: Fig. 6 and Table II, respectively, show the subjective and objective comparison results of the CSDBF and compared methods. As shown in Fig. 6, the false alarm rate of PBCNN, DSFA, and GETNET is too high. Meanwhile, DBDA and SSCNN-S also have some large areas of false alarms. Although SFBS-FFGNET has fewer false alarms than other methods, it has the highest missed alarm rate. Both the false alarm rate and the missed alarm rate of CSDBF

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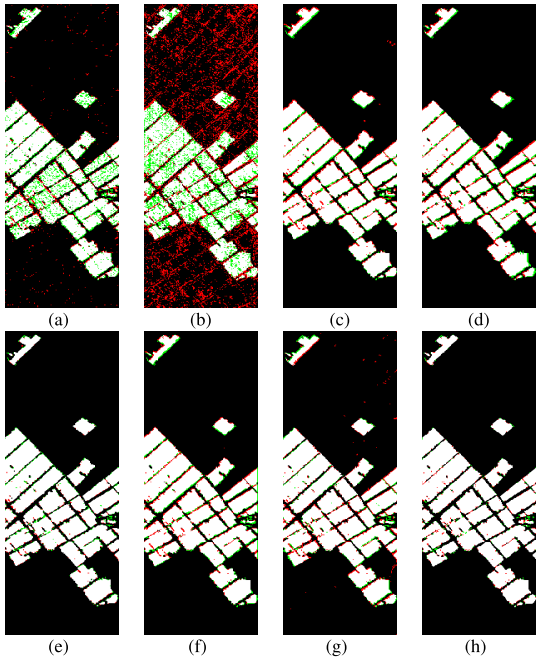


Fig. 7. CD results of different methods on the Farmland dataset (the missed alarms and false alarms are marked in green and red, respectively). (a) PBCNN. (b) DSFA. (c) GETNET. (d) HybridSN. (e) DBDA. (f) SSCNN-S. (g) SFBS-FFGNET. (h) CSDBF.

TABLE III
QUANTITATIVE CD EVALUATIONS ON THE FARMLAND DATASET

Method	OA	Kappa
PBCNN	0.9185	0.7949
DSFA	0.7831	0.5061
GETNET	0.9753	0.9394
HybridSN	0.9749	0.9392
DBDA	0.9789	0.9478
SSCNN-S	0.9774	0.9440
SFBS-FFGNET	0.9760	0.9430
CSDBF	**0.9843±0.0007**	**0.9620±0.0017**
	↑0.54%	**↑1.42%**

are significantly lower than other methods. As can be seen from Table II, due to the class imbalance in this dataset, the OA values of most methods are similar but there are significant differences in Kappa. The proposed CSDBF has obvious advantages in the Kappa index and achieves a 2.72% improvement over the second-place SFBS-FFGNET.

b) Farmland dataset: Fig. 7 and Table III, respectively, show the subjective and objective comparison results of the CSDBF and compared methods on the Farmland dataset. As shown in Fig. 7, there is excessive error detection in PBCNN and DSFA. For GETNET, HybridSN, SSCNN-S, and SFBS-FFGNET, it can be seen that there are many false alarms at the edge of the changed regions. However, DBDA and CSDBF have good detection results, so it is difficult to judge which method is better from the detection result figure directly. According to Table III, the presentation of objective indicators is consistent with the subjective results. Among these methods, the proposed CSDBF is significantly superior to other methods in two indicators. OA and Kappa are improved by 0.54% and 1.42%, respectively, compared with the second place.

c) USA dataset: Fig. 8 and Table IV, respectively, show the subjective and objective comparison results of the proposed CSDBF and compared methods on the USA dataset. It should be noted that since [27] did not conduct experiments on this

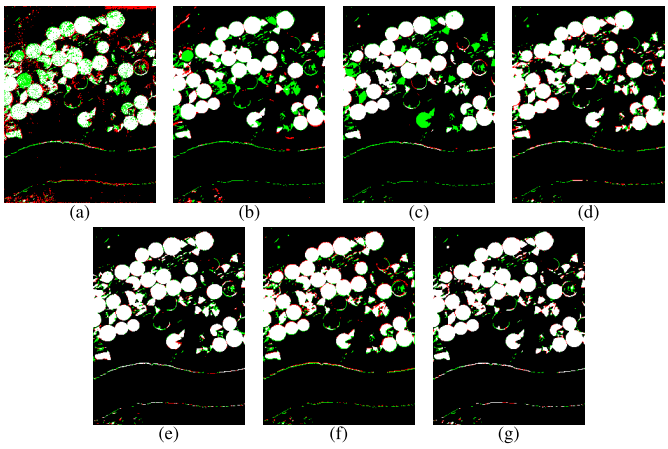


Fig. 8. CD results of different methods on the USA dataset (the missed alarms and false alarms are marked in green and red, respectively). (a) PBCNN. (b) DSFA. (c) GETNET. (d) HybridSN. (e) DBDA. (f) SSCNN-S. (g) CSDBF.

TABLE IV
QUANTITATIVE CD EVALUATIONS ON THE USA DATASET

Method	OA	Kappa
PBCNN	0.8902	0.6699
DSFA	0.9490	0.8430
GETNET	0.9430	0.8249
HybridSN	0.9553	0.8701
DBDA	0.9631	0.8911
SSCNN-S	0.9651	0.8918
CSDBF	**0.9687±0.0008**	**0.9093±0.0024**
	↑0.36%	**↑1.75%**

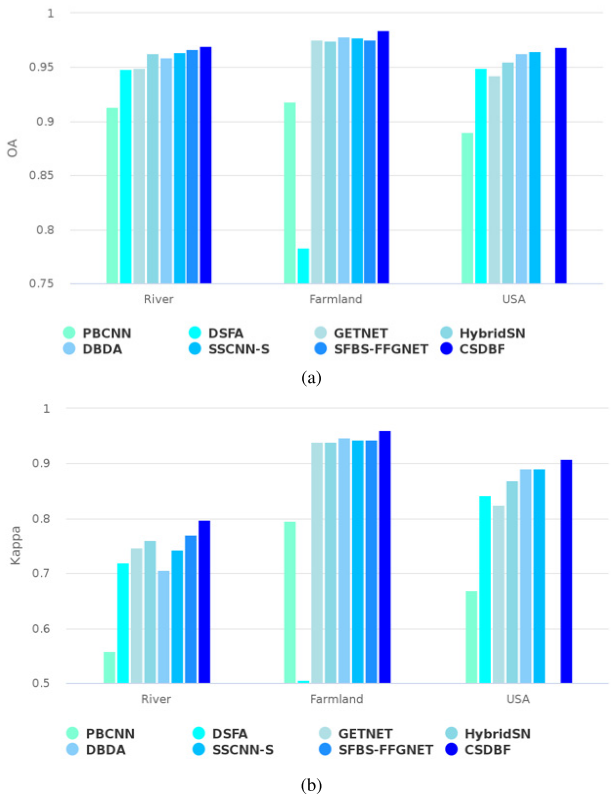


Fig. 9. Objective evaluation indices of different methods in the first group. (a) OA index. (b) Kappa index.

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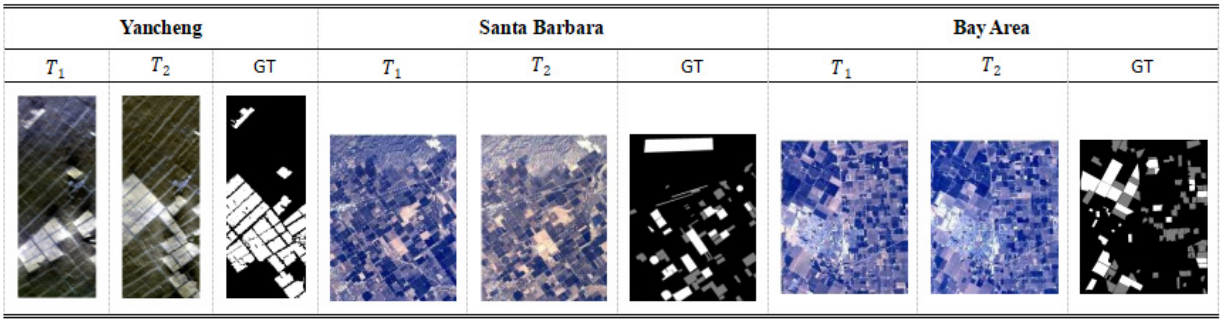


Fig. 10. Pseudocolor images and ground-truth images of the datasets in the second group of experiments.

TABLE V
DATASET INFORMATION

Dataset	Yancheng	Santa Barbara	Bay Area
Size	420×140	984×740	600×500
Spectral Bands	154	224	224
Date T_1	May 3rd, 2006	2013	2013
Date T_2	Apr 23rd, 2007	2014	2015
Sensor	Hyperion	AVIRIS	AVIRIS

dataset, the experimental results of SFBS-FFGNET on this dataset are not listed in this article to reflect fairness. As can be seen from the subjective results shown in Fig. 8, it is obvious that PBCNN, DSFA, and GETNET have high missed alarm rates. Compared with several other methods, the proposed CSDBF is more accurate in capturing river line changes and has a better detection effect for very fine changes. As shown in Table IV, only the OA values of DBDA, SSCNN-S, and CSDBF method exceed 0.96, and only the Kappa value of CSDBF exceeds 0.90, which is 1.75% higher than that of the second place.

d) Summary of this group of experiments: To compare the objective performance of different CD methods more intuitively, Fig. 9(a) and (b), respectively, show the intuitive performance of different methods in OA and Kappa indices for the three datasets. It is clear that the proposed CSDBF has achieved the highest OA and Kappa coefficients in all the three datasets, and the Kappa coefficients have been greatly improved.

B. Second Group of Experiments: Low Sampling Rate

1) Dataset Description: This group of experiments conducts on three public real-world HSI-CD datasets: Yancheng [8], Santa Barbara [61], and Bay Area [61]. The details of the three datasets are shown in Table V. Fig. 10 shows the pseudocolor images and ground-truth images. Especially, for the Santa Barbara and Bay Area datasets, the white pixels, gray pixels, and black pixels in the ground-truth images denote the unchanged regions, changed regions, and unmarked regions, respectively (the CD results of different methods are also represented in this way). Only unchanged pixels and changed pixels are used for training and testing.

2) Sampling Rate and Compared Methods: This group of experiments adopts the same sampling rate as the literature [49]. That is, 0.5% of the samples are randomly selected for training in the Yancheng, Santa Barbara, and Bay Area datasets, while the remaining labeled samples are used for validation. Besides, in our experiments, the segmentation scales of the Yancheng, Santa Barbara, and Bay Area datasets are set to 15, 450, and 80, respectively. Some widely used and advanced approaches are selected for comparative analysis in experiments, including GETNET [25], HSI bidirectional encoder representation from transformers (HSI-BERT) [32], HybridSN [59], recurrent convolution neural network (ReCNN) [62], spectral–spatial residual network (SSRN) [63], BCNNs [28], and D^{2}AGCN [49].

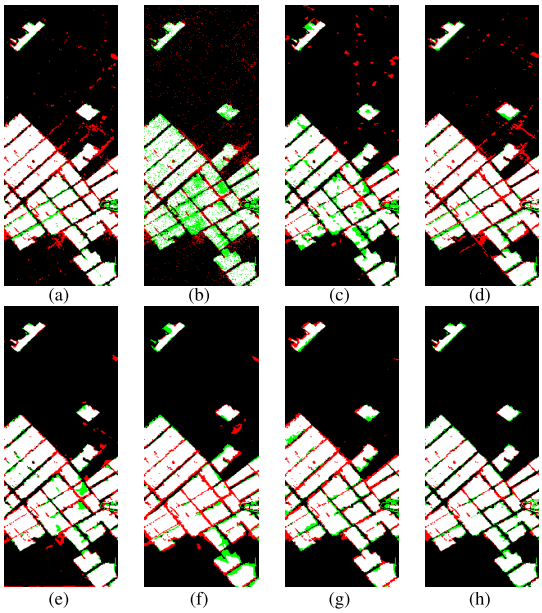


Fig. 11. CD results of different methods on the Yancheng dataset (the missed alarms and false alarms are marked in green and red, respectively). (a) GETNET. (b) HSI-BERT. (c) HybridSN. (d) ReCNN. (e) SSRN. (f) BCNNs. (g) D^{2}AGCN. (h) CSDBF.

3) Analysis of Experimental Results:

a) Yancheng dataset: Fig. 11 and Table VI show the subjective and objective comparison results of the proposed CSDBF and compared methods on the Yancheng dataset. As can be seen from Fig. 11, the HSI-BERT, HybridSN, and BCNN methods have a high rate of missed alarms. The detection results of GETNET, HSI-BERT, HybridSN, and ReCNN contain too many false alarms. Among these methods, the proposed CSDBF is low in both false alarm rate and missed alarm rate. It can be seen from Table VI that the proposed

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TABLE VI

QUANTITATIVE CD EVALUATIONS ON THE YANCHENG DATASET

Method	OA	Kappa
GETNET	0.9371	0.8563
HSI-BERT	0.8765	0.7037
HybridSN	0.9177	0.8073
ReCNN	0.9287	0.8384
SSRN	0.9328	0.8465
BCNNs	0.9331	0.8470
D^2AGCN	0.9374	0.8568
CSDBF	**0.9455±0.0024**	**0.8731±0.0058**
	↑0.81%	**↑1.63%**

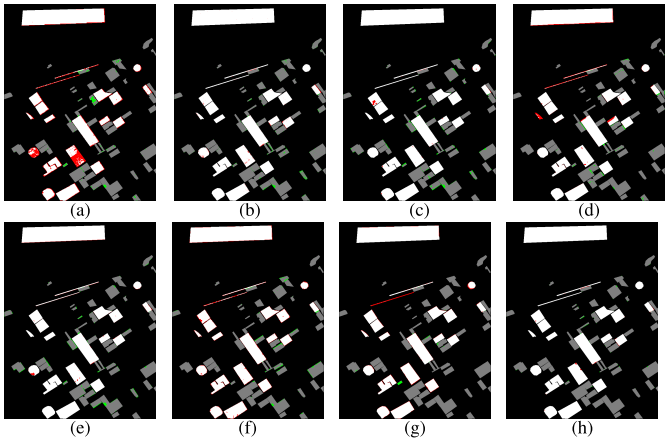


Fig. 12. CD results of different methods on the Santa Barbara dataset (the missed alarms and false alarms are marked in green and red, respectively). (a) GETNET. (b) HSI-BERT. (c) HybridSN. (d) ReCNN. (e) SSRN. (f) BCNNs. (g) D^2AGCN. (h) CSDBF.

TABLE VII

QUANTITATIVE CD EVALUATIONS ON THE SANTA BARBARA DATASET

Method	OA	Kappa
GETNET	0.9126	0.8152
HSI-BERT	0.9811	0.9605
HybridSN	0.9774	0.9525
ReCNN	0.9730	0.9433
SSRN	0.9803	0.9586
BCNNs	0.9762	0.9510
D^2AGCN	0.9803	0.9588
CSDBF	**0.9925±0.0008**	**0.9843±0.0016**
	↑1.14%	**↑2.38%**

CSDBF is superior to other methods in these two indicators. OA and Kappa are 0.81% and 1.63% higher than the second place, respectively.

b) Santa Barbara dataset: Fig. 12 and Table VII show the subjective and objective comparison results of the proposed CSDBF and compared methods on the Santa Barbara dataset. As can be seen from Fig. 12, GETNET and ReCNN have a high false alarm rate. Besides, HybridSN, SSRN, and BCNNs have a high missed alarm rate. HSI-BERT, D^2AGCN, and the proposed CSDBF perform well. In Table VII, the proposed CSDBF is significantly superior to other methods in the two indices. Compared with the second-ranked method HSI-BERT, the CSDBF method has improved by 1.14% and 2.38% in OA and Kappa indices, respectively.

c) Bay area dataset: Fig. 13 and Table VIII show the subjective and objective comparison results of the proposed CSDBF and compared methods on the Bay Area dataset. As can be seen from the subjective results shown in

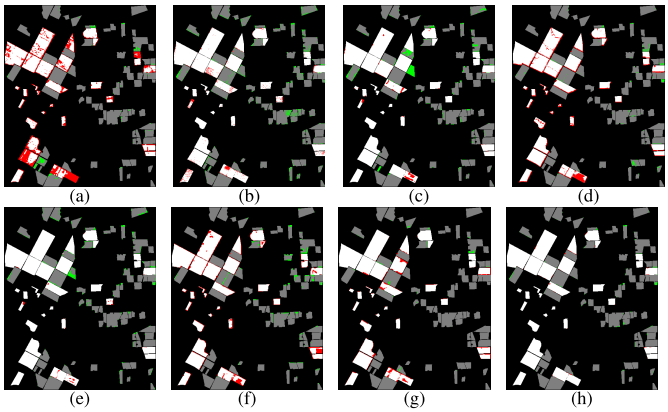


Fig. 13. CD results of different methods on the Bay Area dataset (the missed alarms and false alarms are marked in green and red, respectively). (a) GETNET. (b) HSI-BERT. (c) HybridSN. (d) ReCNN. (e) SSRN. (f) BCNNs. (g) D^2AGCN. (h) CSDBF.

TABLE VIII

QUANTITATIVE CD EVALUATIONS ON THE BAY AREA DATASET

Method	OA	Kappa
GETNET	0.8687	0.7564
HSI-BERT	0.9460	0.8917
HybridSN	0.9539	0.9073
ReCNN	0.9526	0.9048
SSRN	0.9664	0.9327
BCNNs	0.9604	0.9210
D^2AGCN	0.9690	0.9376
CSDBF	**0.9871±0.0021**	**0.9739±0.0042**
	↑1.81%	**↑3.63%**

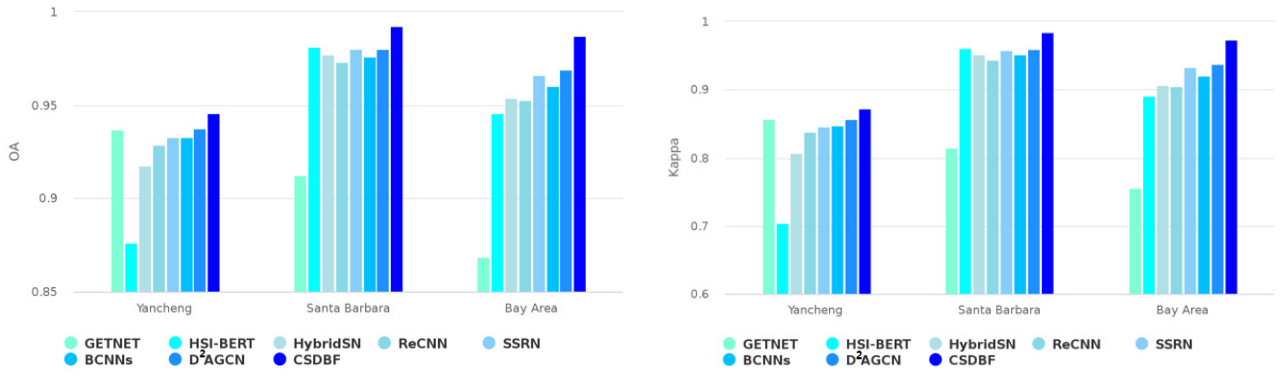
Fig. 13, GETNET and ReCNN have a high false alarm rate. Differently, there are many missed alarm phenomena in the results of HSI-BERT, HybridSN, SSRN, and BCNNs. Although D^2AGCN has a competitive performance, it still has more false alarms than the proposed CSDBF in this article. As can be seen from Table VIII, the proposed CSDBF method is ahead of other methods in these two indices and has advantages of 1.81% and 3.63% in OA and Kappa indices compared with D^2AGCN, respectively, which ranks second.

d) Summary of this group of experiments: To compare the objective performance of different CD methods more intuitively, Fig. 14(a) and (b), respectively, shows the intuitive performance of different methods in OA and Kappa indices on the three datasets. The CSDBF has achieved the highest OA and Kappa coefficients in all the three datasets, and the Kappa coefficients were increased greatly.

C. Effectiveness Study on TSJGAT and Ablation Study on Experiment of CSDBF

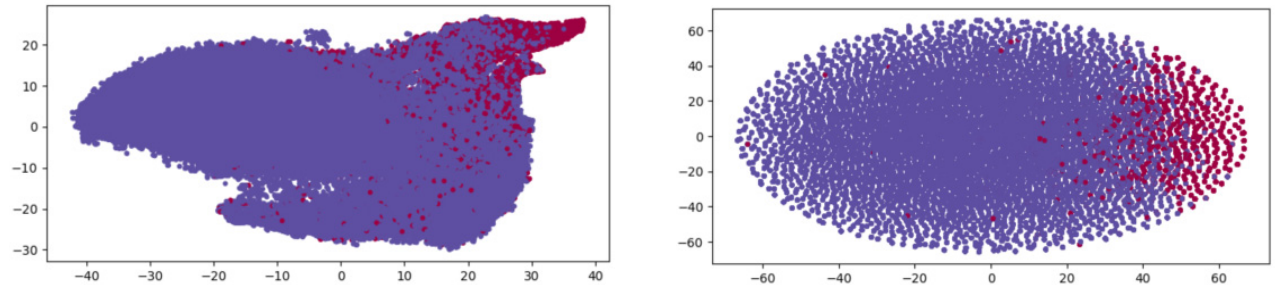
1) Effectiveness Study on TSJGAT: The proposed TSJGAT realizes the aggregation of features in the graph domain in the form of nodes and effectively captures the nonlinear properties of bitemporal HSIs, which lays a foundation for effective detection of complex ground object changes in the images. To illustrate the effectiveness of this module, Fig. 15 takes the experiment of the River dataset under the condition of a 0.1% sampling rate as an example and presents the t-distributed stochastic neighbor embedding (t-SNE) visual display of feature maps before and after the implementation of

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(a) (b)

Fig. 14. Objective evaluation indices of different methods in the second group. (a) OA index. (b) Kappa index.



(a) (b)

Fig. 15. T-SNE visualizations of the feature maps output before and after implementing TSJGAT. (a) Before implementing TSJGAT. (b) After implementing TSJGAT.

TABLE IX
OA AND KAPPA INDICES OF DIFFERENT BRANCHES

Dataset	Index	P-branch	S-branch	CSDBF
River	OA	0.9686	0.9549	**0.9697**
	Kappa	0.7894	0.7038	**0.7982**
Farmland	OA	0.9820	0.9669	**0.9843**
	Kappa	0.9564	0.9193	**0.9620**
USA	OA	0.9680	0.9424	**0.9687**
	Kappa	0.9074	0.8288	**0.9093**
Yancheng	OA	0.9256	0.9270	**0.9455**
	Kappa	0.8357	0.8311	**0.8731**
Santa Barbara	OA	**0.9927**	0.9836	0.9925
	Kappa	**0.9849**	0.9659	0.9843
Bay Area	OA	0.9832	0.9761	**0.9871**
	Kappa	0.9662	0.9519	**0.9739**

TSJGAT. In Fig. 15, red and purple feature clouds represent the distribution of changed and unchanged features, respectively. The feature distribution after the implementation of TSJGAT obviously has better separability than the previous feature distribution. Thus, in the case of low sampling rates, TSJGAT can enhance the feature expression ability of CSDBF for details and class boundaries, and eliminate many fake changes.

2) Ablation Studies on CSDBF:

a) Ablation experiments of branches: There are two branches in the proposed CSDBF (see Fig. 1), namely, the superpixel-level feature extraction branch (S-branch) and pixel-level feature extraction branch (P-branch). To evaluate their effectiveness, ablation experiments based on these two branches are designed. Table IX presents the objective metrics of ablation experiments on six datasets with the same sampling rate as the comparison experiments above.

It can be seen from Table IX that the results of parallel detection of the P-branch and S-branch are superior to independent detection of the two branches in most cases, indicating that the complementarity of the proposed framework CSDBF exists, and it is effective and necessary. However, on the Santa Barbara dataset, the CD performance of the P-branch is better than that of CSDBF. This is because the scales of the changed regions in this dataset are generally very large, while the segmentation scale we selected for this dataset is too small, resulting in too much noise in the graph we constructed. Thus, the CD performance of the S-branch is greatly affected, and the complementarity between the S-branch and P-branch cannot be played. These reasons can also be proved by the ablation experiments on the superpixel segmentation scale in Section IV-C2.c.

b) Ablation experiments on the complementarity of branches: To illustrate the complementary working mode of the P-branch and S-branch in the network, and the advantages of the S-branch in the case of small numbers of samples, ablation experiments concerning sampling rates on the Farmland and River datasets are further provided. We select the proportions of 0.1%, 0.5%, 1%, 5%, 10%, and 20% samples for training in the ablation experiments to analyze the detection performance of the two branches under different conditions. The experimental results are shown in Figs. 16 and 17.

It is easy to find from Figs. 16 and 17 that when the sampling rates are very low, such as 0.1% and 0.5%, the P-branch has a poor independent detection effect and too many false alarms. However, the S-branch can use the graph structure to spread the knowledge to the place without labels,

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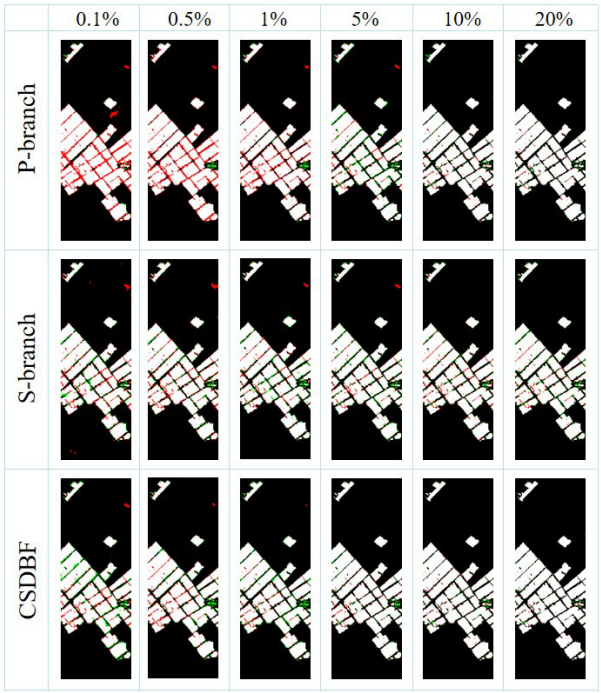


Fig. 16. CD results of different sampling rates of training samples and different branches on the Farmland dataset (the missed alarms and false alarms are marked in green and red, respectively).

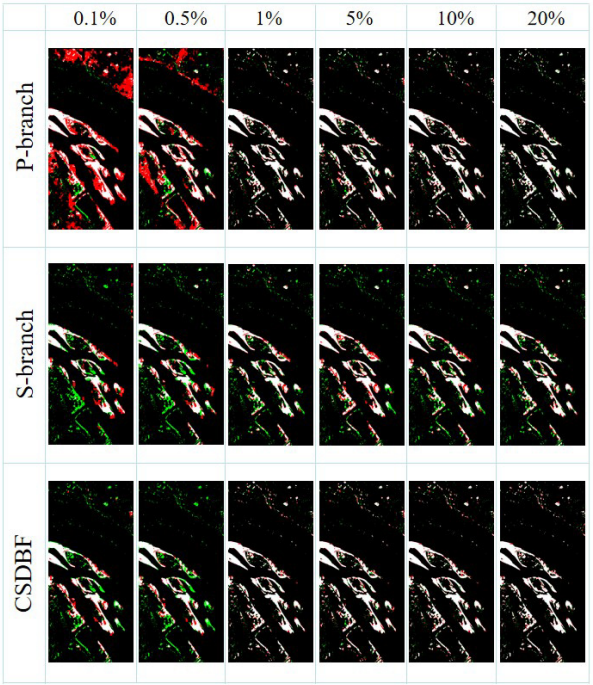


Fig. 17. CD results of different sampling rates of training samples and different branches on the River dataset (the missed alarms and false alarms are marked in green and red, respectively).

so as to obtain more accurate detection. Meanwhile, as the sampling rates increase, the independent detection effect of the P-branch is significantly improved, while the detection results of the S-branch show little improvement. Thus, when the sampling rate is high, the P-branch plays a major role, while the S-branch plays a complementary role. Furthermore,

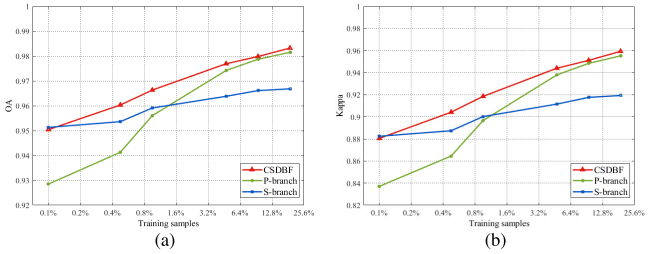


Fig. 18. OA and Kappa of different branch combinations on the Farmland dataset under different sampling rates. (a) OA. (b) Kappa.

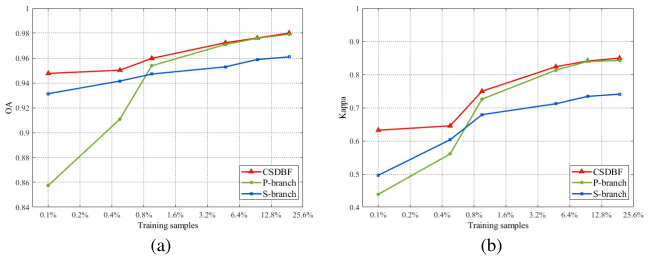


Fig. 19. OA and Kappa of different branch combinations on the River dataset under different sampling rates. (a) OA. (b) Kappa.

Tables X and XI show OA and Kappa indices of the ablation experiment results. To make the experimental results more intuitive, Figs. 18 and 19 show the broken line graphs corresponding to Tables X and XI.

It can be seen from Figs. 18 and 19, Tables X, and XI that the detection synergy of CSDBF is consistent with that observed in Figs. 16 and 17. That is, when the sampling rate is low, the S-branch plays a major role and its detection performance is better than that of the P-branch because the TSJGAT in the P-branch effectively makes use of the unlabeled information. However, when the sampling rate is high, the P-branch plays a major role and its detection performance is better than the S-branch because CNN performs pixel-level feature extraction and corrects error accumulation in the superpixel segmentation module. Therefore, in this article, the two branches are put into a network in a cooperative detection mode, so that the network can give full play to the complementary detection performance of the parallel dual branches. Meanwhile, by analyzing the variation trend of the P-branch and S-branch as the number of samples increases, it can be found that the sensitivity of the S-branch to the number of training samples is significantly lower than that of the P-branch, indicating that the S-branch is more stable and suitable for more HSI-CD tasks.

c) Influence of segmentation scale: The superpixel segmentation scale c is an important hyperparameter and could affect the performance of CSDBF. To analyze its influence, we select multiple segmentation scales for each dataset to conduct experiments in this section. Each experiment is repeated five times, and the means and standard deviations of OA are shown in Fig. 20.

The segmentation scale affects the size of the graph structure. The larger the segmentation scale, the smaller the graph structure and computational complexity. Many fine changes may be completely ignored, but the noise can be effectively suppressed. On the contrary, small segmentation scales lead to

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TABLE X

OA AND KAPPA INDICES OF DIFFERENT BRANCHES ON THE FARMLAND DATASET

Branch	Index	Training samples (total rate, positive samples, negative samples)					
		0.1%	0.5%	1%	5%	10%	20%
		18, 45	91, 224	183, 447	914, 2236	1827, 4473	3654, 8946
P-branch	OA	0.9285	0.9413	0.9560	0.9742	0.9787	0.9815
	Kappa	0.8371	0.8645	0.8966	0.9379	0.9484	0.9552
S-branch	OA	**0.9513**	0.9536	0.9591	0.9638	0.9661	0.9668
	Kappa	**0.8824**	0.8874	0.9002	0.9115	0.9176	0.9193
CSDBF	OA	0.9504	**0.9603**	**0.9663**	**0.9769**	**0.9798**	**0.9832**
	Kappa	0.8805	**0.9042**	**0.9185**	**0.9440**	**0.9511**	**0.9593**

TABLE XI

OA AND KAPPA INDICES OF DIFFERENT BRANCHES ON THE RIVER DATASET

Branch	Index	Training samples (total rate, positive samples, negative samples)					
		0.1%	0.5%	1%	5%	10%	20%
		19, 93	93, 465	186, 930	930, 4650	1860, 9298	3719, 18597
P-branch	OA	0.8575	0.9109	0.9538	0.9709	0.9759	0.9791
	Kappa	0.4388	0.5606	0.7260	0.8132	0.8398	0.8422
S-branch	OA	0.9313	0.9415	0.9471	0.9529	0.9588	0.9609
	Kappa	0.4963	0.6035	0.6791	0.7119	0.7341	0.7408
CSDBF	OA	**0.9476**	**0.9501**	**0.9597**	**0.9723**	**0.9761**	**0.9801**
	Kappa	**0.6322**	**0.6452**	**0.7498**	**0.8236**	**0.8410**	**0.8495**

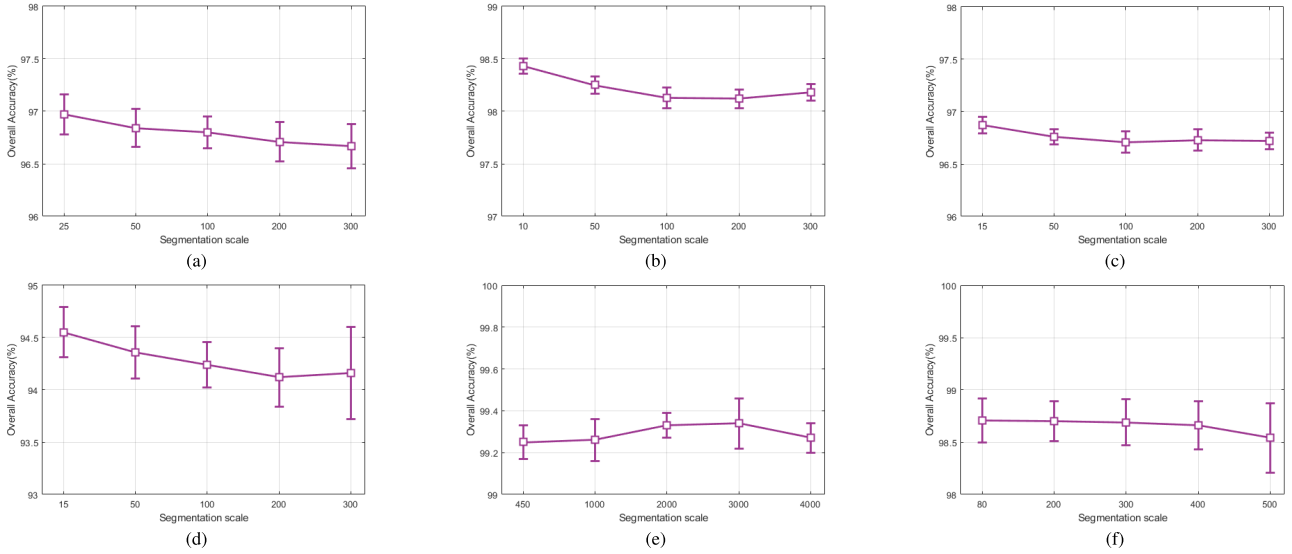


Fig. 20. CD performance of CSDBF with different segmentation scales on each dataset. (a) River. (b) Farmland. (c) USA. (d) Yancheng. (e) Santa Barbara. (f) Bay Area.

larger graph structures and computational complexity but make it possible to capture subtle changes. According to the results, the accuracies of CSDBF on the River, Farmland, USA, Yancheng, and Bay Area datasets decrease with the increase in scales, which indicates that there are more subtle changes in these datasets. Differently, in the Santa Barbara dataset, when the segmentation scale increases, the accuracy continues to rise and finally decreases, which shows that this dataset contains more large-scale changes and less subtle changes. When the scale reaches more than 4000, the segmentation scale exceeds the area of many large-scale changed regions, resulting in much incorrect segmentation and too much noise in the graph. So the accuracy starts to go down. Considering the detection accuracy and computational complexity, the scales we chose in the first group of experiments are 25, 10, and 15, respectively. For the sake of comparison, the scales we chose in the second group of experiments are 15, 450, and 80, respectively, which are the same as those set in [49].

d) Comparison of conditional random field and TSJGAT: Conditional random field (CRF) is considered an effective method to enhance the edge continuity of classification results or semantic segmentation results and eliminate holes or disconnected regions in the classification map obtained by CNNs [64]. Since GNNs have many similar functions to CRF in these aspects, we conduct a comparative experiment on CRF and TSJGAT by referring to the experiments in [65]. The River, Farmland, and USA datasets are selected for this experiment. Besides, we conduct experiments at both low and high sampling rates to facilitate a more comprehensive comparison. The PRE results and evaluation indices are shown in Fig. 21 and Table XII.

It can be seen that OA and Kappa of CRF and CSDBF are improved compared with those of the P-branch. However, the advantages of CRF and CSDBF are quite different. Since CRF is a postprocessing method to improve the results of the P-branch and aims at reducing noise and enhancing

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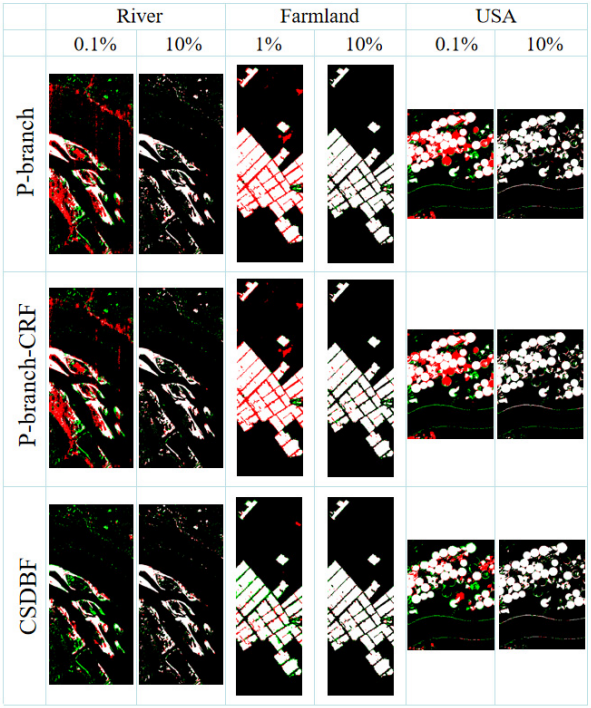


Fig. 21. CD results of different sampling rates of training samples and different branches on the River dataset (the missed alarms and false alarms are marked in green and red, respectively).

TABLE XII
OA AND KAPPA INDICES OF DIFFERENT METHODS

Dataset	Samples	Index	P-branch	P-branch-CRF	CSDBF
River	0.1%	OA	0.8575	0.8610	**0.9476**
		Kappa	0.4388	0.4468	**0.6322**
	10%	OA	0.9759	**0.9767**	0.9761
		Kappa	0.8398	0.8403	**0.8410**
Farmland	0.1%	OA	0.9285	0.9285	**0.9504**
		Kappa	0.8371	0.8370	**0.8805**
	10%	OA	0.9787	0.9797	**0.9798**
		Kappa	0.9484	**0.9524**	0.9511
USA	0.1%	OA	0.8849	0.8850	**0.8969**
		Kappa	0.6873	0.6875	**0.7060**
	10%	OA	0.9688	**0.9713**	0.9691
		Kappa	0.9079	**0.9104**	0.9095
Increase Rate		OA	-	↑0.13%	**↑1.34%**
		Kappa	-	↑0.25%	**↑4.35%**

TABLE XIII
TOTAL PARAMETERS AND RUNNING TIMES OF DIFFERENT METHODS ON THE RIVER DATASET

Method	PBCNN	GETNET	DBDA	SSCNN-S	HSI-BERT	SSRN	HybridSN	CSDBF
Params(k)	235	154179	377	22	225	361	402	142
Running Time(s)	557	421	784	197	867	1286	186	394

continuity, the PRE maps obtained by CRF are very similar to those of the P-branch. So OA and Kappa indices of the P-branch-CRF are always very close to those of the P-branch, no matter whether the accuracies of the P-branch are high or low. Differently, CSDBF not only improves the accuracies compared with the P-branch when the sampling rate is high but also shows great potential when the sampling rate is low. In conclusion, TSJGAT has more advantages than CRF when the number of training samples is small and can be applied to more HSI-CD tasks.

D. Time Efficiency Analysis of Different Models

In addition to accuracy, time efficiency is also a very important index to measure the effectiveness of the methods. Considering that the comparison algorithms in the two groups of experiments are different, we select some models from each group and conducted experiments on the River dataset to compare the efficiency of different methods fairly. The numbers of parameters and the total running time of different methods are shown in Table XIII. As shown in Table XIII, both the number of parameters and the running time are smaller than most methods, which indicates that CSDBF can achieve competitive time efficiency.

V. CONCLUSION

In this article, a dual-branch framework based on TSJGAT with a complementary strategy is proposed, which can make full use of the advantages of CNN and GNN by two branches and make the two branches work together to achieve better detection performance than themselves. In the superpixel-level feature extraction branch, we introduce the idea of GATs into the HSI-CD field for the first time and design a suitable TSJGAT module for the HSI-CD task, which carries out feature propagation using the temporal–spatial joint correlations in the graph domain in the form of nodes. In the pixel-level feature extraction branch, the pixel-level features obtained by naive CNNs can correct the error accumulation caused by superpixel segmentation and form complementary advantages with the superpixel-level feature extraction branch. A large number of comparison and ablation experiments on six commonly used datasets demonstrate the superior property of the proposed CSDBF.

REFERENCES

[1] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.
[2] J. Choi, B. Wang, G. Kim, and Y. Han, "Hyperspectral change detection by using IR-MAD and synthetic image fusion," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 1658–1661.
[3] Q. Guo, J. Zhang, and Y. Zhang, "Multitemporal hyperspectral images change detection based on joint unmixing and information coguidance strategy," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9633–9645, Nov. 2021.
[4] F. Huang, Y. Yu, and T. Feng, "Hyperspectral remote sensing image change detection based on tensor and deep learning," *J. Vis. Commun. Image Represent.*, vol. 58, pp. 233–244, Jan. 2019.
[5] B. Taskesen, A. Koz, A. Alatan, and O. Weatherbee, "Change detection for hyperspectral images using extended mutual information and oversegmentation," in *Proc. 9th Workshop Hyperspectral Image Signal Process., Evol. Remote Sens. (WHISPERS)*, Sep. 2018, pp. 1–5.
[6] S. Liu, D. Marinelli, L. Bruzzone, and F. Bovolo, "A review of change detection in multitemporal hyperspectral images: Current techniques, applications, and challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 140–158, Jun. 2019.
[7] J. Dhawan and P. Narula, "An exploration of various change detection techniques for hyperspectral images," *IPASJ Int. J. Comput. Sci.*, vol. 3, no. 3, pp. 26–29, 2015.
[8] M. Hansanlou and S. T. Seydi, "Hyperspectral change detection: An experimental comparative study," *Remote Sens.*, vol. 39, no. 20, pp. 7029–7083, Oct. 2018.

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[9] W. A. Malila, "Change vector analysis: An approach for detecting forest changes with Landsat," in *Proc. LARS Symposia*, 1980, p. 385.

[10] D. Lu, P. Mausel, E. Brondizio, and E. Moran, "Change detection techniques," *Int. J. Remote Sens.*, vol. 25, no. 12, pp. 2365–2401, Apr. 2004.

[11] J. S. Deng, K. Wang, Y. H. Deng, and G. Qi, "PCA-based land-use change detection and analysis using multitemporal and multisensor satellite data," *Int. J. Remote Sens.*, vol. 29, no. 16, pp. 4823–4838, 2008.

[12] A. A. Nielsen, "The regularized iteratively reweighted mad method for change detection in multi-and hyperspectral data," *IEEE Trans. Image Process.*, vol. 16, no. 2, pp. 463–478, Feb. 2007.

[13] F. Bovolo, L. Bruzzone, and M. Marconcini, "A novel approach to unsupervised change detection based on a semisupervised SVM and a similarity measure," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 7, pp. 2070–2082, Jul. 2008.

[14] R. Mousazadeh, H. Ghaffarzadeh, J. Nouri, A. Gharagozlou, and M. Farahpour, "Land use change detection and impact assessment in Anzali international coastal wetland using multi-temporal satellite images," *Environ. Monitor. Assessment*, vol. 187, no. 12, pp. 1–11, Dec. 2015.

[15] Y. Afaq and A. Manocha, "Analysis on change detection techniques for remote sensing applications: A review," *Ecol. Informat.*, vol. 63, Jul. 2021, Art. no. 101310, doi: 10.1016/J.ECOINF.2021.101310.

[16] T. Kasetkasem and P. K. Varshney, "An image change detection algorithm based on Markov random field models," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 8, pp. 1815–1823, Aug. 2002.

[17] J. Meola, M. T. Eismann, R. L. Moses, and J. N. Ash, "Application of model-based change detection to airborne VNIR/SWIR hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 10, pp. 3693–3706, Oct. 2012.

[18] A. Erturk, S. Erturk, and A. Plaza, "Unmixing with SLIC superpixels for hyperspectral change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 3370–3373.

[19] S. Liu, L. Bruzzone, F. Bovolo, and P. Du, "Unsupervised multitemporal spectral unmixing for detecting multiple changes in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 5, pp. 2733–2748, May 2016.

[20] S. Zhang et al., "Superpixel-guided sparse unmixing for remotely sensed hyperspectral imagery," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 2155–2158.

[21] L. Khelifi and M. Mignotte, "Deep learning for change detection in remote sensing images: Comprehensive review and meta-analysis," *IEEE Access*, vol. 8, pp. 126385–126400, 2020.

[22] J. Lopez-Fandino, A. S. Garea, D. B. Heras, and F. Arguello, "Stacked autoencoders for multiclass change detection in hyperspectral images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 1906–1909.

[23] M. Papadomanolaki, S. Verma, M. Vakalopoulou, S. Gupta, and K. Karantzalos, "Detecting urban changes with recurrent neural networks from multitemporal Sentinel-2 data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 214–217.

[24] M. Gong, J. Zhao, J. Liu, Q. Miao, and L. Jiao, "Change detection in synthetic aperture radar images based on deep neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 1, pp. 125–138, Jan. 2015.

[25] Q. Wang, Z. Yuan, Q. Du, and X. Li, "GETNET: A general end-to-end 2-D CNN framework for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 3–13, Jan. 2018.

[26] T. Zhan et al., "A spectral–spatial convolution neural network with Siamese architecture for change detection," *Remote Sens.*, vol. 13, no. 5, pp. 895–912, 2021.

[27] X. Ou, L. Liu, B. Tu, G. Zhang, and Z. Xu, "A CNN framework with slow-fast band selection and feature fusion grouping for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

[28] Y. Lin, S. Li, L. Fang, and P. Ghamisi, "Multispectral change detection with bilinear convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 10, pp. 1757–1761, Oct. 2020.

[29] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

[30] J. Bai, B. Ding, Z. Xiao, L. Jiao, H. Chen, and A. C. Regan, "Hyperspectral image classification based on deep attention graph convolutional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

[31] S. Baroud, S. Chokri, S. Belhaous, and M. Mestari, "A brief review of graph convolutional neural network based learning for classifying remote sensing images," *Proc. Comput. Sci.*, vol. 191, pp. 349–354, Jan. 2021.

[32] J. He, L. Zhao, H. Yang, M. Zhang, and W. Li, "HSI-BERT: Hyperspectral image classification using the bidirectional encoder representation from transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 165–178, Sep. 2019.

[33] X. He, Y. Chen, and Z. Lin, "Spatial–spectral transformer for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 3, p. 498, Jan. 2021.

[34] Y. Wang et al., "Spectral–spatial–temporal transformers for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022, doi: 10.1109/TGRS.2022.3203075.

[35] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[36] X. Yang, W. Cao, Y. Lu, and Y. Zhou, "Hyperspectral image transformer classification networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[37] L. Waikhom and R. Patgiri, "Graph neural networks: Methods, applications, and opportunities," 2021, *arXiv:2108.10733*.

[38] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and S. Y. Philip, "A comprehensive survey on graph neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 1, pp. 4–24, Mar. 2020.

[39] R. Singh, S. Bathla, and P. Meel, "State-of-the-art applications of graph convolutional neural networks," in *Proc. 6th Int. Conf. Recent Trends Comput.* Cham, Switzerland: Springer, 2021, pp. 107–115.

[40] F. Scarselli, A. C. Tsoi, M. Gori, and M. Hagenbuchner, "Graphical-based learning environments for pattern recognition," in *Proc. Joint IAPR Int. Workshops Stat. Techn. Pattern Recognit. (SPR) Struct. Syntactic Pattern Recognit. (SSPR)*. Cham, Switzerland: Springer, 2004, pp. 42–56.

[41] F. Scarselli, M. Gori, A. C. Tsoi, M. Hagenbuchner, and G. Monfardini, "The graph neural network model," *IEEE Trans. Neural Netw.*, vol. 20, no. 1, pp. 61–80, Dec. 2009.

[42] J. Bruna, W. Zaremba, A. Szlam, and Y. LeCun, "Spectral networks and locally connected networks on graphs," 2013, *arXiv:1312.6203*.

[43] M. Defferrard, X. Bresson, and P. Vandergheynst, "Convolutional neural networks on graphs with fast localized spectral filtering," in *Proc. 30th Int. Conf. Neural Inf. Process. Syst. (NIPS)*. Barcelona, Spain, 2016, pp. 3844–3852.

[44] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," 2016, *arXiv:1609.02907*.

[45] Q. Liu, L. Xiao, J. Yang, and Z. Wei, "CNN-enhanced graph convolutional network with pixel-and superpixel-level feature fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8657–8671, Oct. 2020.

[46] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Lio, and Y. Bengio, "Graph attention networks," 2017, *arXiv:1710.10903*.

[47] A. Sha, B. Wang, X. Wu, and L. Zhang, "Semisupervised classification for hyperspectral images using graph attention networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 1, pp. 157–161, Jan. 2021.

[48] K. Xu, Y. Zhao, L. Zhang, C. Gao, and H. Huang, "Spectral–spatial residual graph attention network for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[49] J. Qu, Y. Xu, W. Dong, Y. Li, and Q. Du, "Dual-branch difference amplification graph convolutional network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022.

[50] D. He, Y. Zhong, and L. Zhang, "Land cover change detection based on spatial–temporal sub-pixel evolution mapping: A case study for urban expansion," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 1970–1973.

[51] R. Song, W. Ni, W. Cheng, and X. Wang, "CSANet: Cross-temporal interaction symmetric attention network for hyperspectral image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2022.3179134.

[52] C. Shi, Z. Zhang, W. Zhang, C. Zhang, and Q. Xu, "Learning multiscale temporal–spatial–spectral features via a multipath convolutional LSTM neural network for change detection with hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022, doi: 10.1109/TGRS.2022.3176642.

[53] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2012.

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[54] V. Jampani, D. Sun, M.-Y. Liu, M.-H. Yang, and J. Kautz, "Superpixel sampling networks," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 352–368.

[55] Q. Liu, L. Xiao, F. Liu, and J. Huan, "SSCDenseNet: A spectral-spatial convolutional dense network for hyperspectral image classification," (in Chinese), *Acta Electronica Sinica*, vol. 48, no. 4, pp. 751–762, Nov. 2020.

[56] Y. Yuan, H. Lv, and X. Lu, "Semi-supervised change detection method for multi-temporal hyperspectral images," *Neurocomputing*, vol. 148, pp. 363–375, Jan. 2015.

[57] A. Sharma, X. Liu, X. Yang, and D. Shi, "A patch-based convolutional neural network for remote sensing image classification," *Neural Netw.*, vol. 95, pp. 19–28, Nov. 2017.

[58] B. Du, L. Ru, C. Wu, and L. Zhang, "Unsupervised deep slow feature analysis for change detection in multi-temporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9976–9992, Dec. 2019.

[59] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[60] R. Li, S. Zheng, C. Duan, Y. Yang, and X. Wang, "Classification of hyperspectral image based on double-branch dual-attention mechanism network," *Remote Sens.*, vol. 12, no. 3, p. 582, Feb. 2020.

[61] L. Javier, H. Dora B, A. Francisco, and D. Mauro, "GPU framework for change detection in multitemporal hyperspectral images," *Int. J. Parallel Program.*, vol. 47, no. 2, pp. 272–292, 2019.

[62] L. Mou, L. Bruzzone, and X. X. Zhu, "Learning spectral–spatial–temporal features via a recurrent convolutional neural network for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 924–935, Feb. 2019.

[63] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[64] F. I. Alam, J. Zhou, A. W.-C. Liew, X. Jia, J. Chanussot, and Y. Gao, "Conditional random field and deep feature learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1612–1628, Mar. 2019.

[65] Y. Liang, X. Zhao, A. J. X. Guo, and F. Zhu, "Hyperspectral image classification with deep metric learning and conditional random field," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 6, pp. 1042–1046, Jun. 2020.



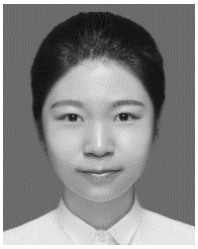
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